I^2SB as posterior ascent with a bridge preconditioner

ImMAP project notes

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Diffusion samplers and bridge samplers do the same thing. Each runs an annealed gradient ascent on a smoothed log-posterior, and a denoiser supplies the gradient. They differ in how the network is shown its input. A very noisy state $\boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ is a poor network input. A bridge re-expresses it as a bridge sample, which carries the same information but stays bounded, so the network never sees large noise.

- Section 1 states the ascent for any side information.
- Section 2 introduces the bridge as a change of variables for the network input, and says why it helps.
- Section 3 derives I^2SB's sampler as this ascent [1], then maps it onto I^2SB's update line by line.

1 Posterior ascent

We want $\max_{\boldsymbol{x}_0} \log p(\boldsymbol{x}_0 \mid \boldsymbol{c})$, where $\boldsymbol{c}$ is the side information: a start image $\boldsymbol{x}_1$, a measurement $\boldsymbol{y}$, or both. As in ImMAP [2], we ascend a Gaussian smoothing of the objective while lowering the smoothing:

$$J_\sigma(\boldsymbol{u}) \coloneqq \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}), \qquad p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}) = \int \mathcal{N}(\boldsymbol{u}; \boldsymbol{x}, \sigma^2 \boldsymbol{I})\, p(\boldsymbol{x} \mid \boldsymbol{c})\, d\boldsymbol{x}.$$

So J_σ is the log-density of the noisy target $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ given $\boldsymbol{c}$, and it tends to the objective as $\sigma \to 0$. Its gradient is a denoiser (Tweedie):

$$\sigma^2 \nabla J_\sigma(\boldsymbol{u}) = D(\boldsymbol{u}, \sigma; \boldsymbol{c}) - \boldsymbol{u}, \qquad D(\boldsymbol{u}, \sigma; \boldsymbol{c}) \coloneqq \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}]. \tag{1}$$

The step. Gradient ascent with step size $h\sigma^2$ is

$$\boldsymbol{u}^+ = \boldsymbol{u} + h\sigma^2 \nabla J_\sigma(\boldsymbol{u}) = (1 - h)\, \boldsymbol{u} + h\, D(\boldsymbol{u}, \sigma; \boldsymbol{c}). \tag{2}$$

Tweedie's second-order identity gives $-\nabla^2 J_\sigma = (\sigma^2 \boldsymbol{I} - \mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}])/\sigma^4 \preceq \sigma^{-2} \boldsymbol{I}$. So every $h \in (0, 2)$ increases J_σ, by the descent lemma.

The continuation. After the step, fresh noise brings the iterate to the next, smaller level $\sigma' = r\sigma$. An exact step leaves noise $(1 - h)\sigma$, so the injection is

$$\gamma^2 = \sigma'^2 - (1 - h)^2 \sigma^2, \qquad \text{which needs } h \in [1 - r,\ 1 + r]. \tag{3}$$

Algorithm 1 Annealed posterior ascent (`debiased_ascent`)

Require: denoiser $D(\cdot, \sigma; \boldsymbol{c})$; ladder $\sigma_N > \cdots > \sigma_1 > \sigma_0 = 0$ ($\sigma_N = \infty$ allowed); steps $h_k \in [1 - r_k, 1]$ with $r_k = \sigma_{k-1}/\sigma_k$

$\boldsymbol{u}_N \leftarrow \sigma_N \boldsymbol{z}$ ▷ any $\boldsymbol{u}_N$ if $\sigma_N = \infty$

for $k = N, \ldots, 1$ **do**

  $\boldsymbol{v}_k \leftarrow D(\boldsymbol{u}_k, \sigma_k; \boldsymbol{c})$ ▷ $= \boldsymbol{u}_k + \sigma_k^2 \nabla J_{\sigma_k}(\boldsymbol{u}_k)$

  $\boldsymbol{u}_{k-1} \leftarrow (1 - h_k)\, \boldsymbol{u}_k + h_k\, \boldsymbol{v}_k + \sqrt{\sigma_{k-1}^2 - (1 - h_k)^2 \sigma_k^2}\; \boldsymbol{z}_k$ ▷ (2), (3)

end for

return $\boldsymbol{u}_0$ ▷ $= \boldsymbol{v}_1$, since $\sigma_0 = 0$ forces $h_1 = 1$

Three step sizes are worth naming:

- $h = 1 - r^2$ removes the variance $\sigma^2 - \sigma'^2$ and injects $\gamma^2 = \sigma'^2(1 - r^2)$. This is the VE ancestral sampler [3].
- $h = 1 - r$ injects nothing: the probability-flow ODE step.
- $h = 1$ jumps to the estimate and re-noises it: SDEdit-style restarts [5].

At $\sigma_N = \infty$ we have $r = 0$, so $h = 1$ is forced. The first step then returns $D(\cdot, \infty; \boldsymbol{c}) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{c}]$, whatever $\boldsymbol{u}_N$ is, and re-noises it to σ_{N-1}: the ascent *warm-starts itself* from the posterior mean.

2 Why a bridge

Algorithm 1 needs the denoiser on the whole ladder, from very large (or infinite) σ down to 0. At large σ the input $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ is almost all noise:

- its norm grows like σ;
- the network must return $\approx \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{c}]$ while discarding all of it.

Standard diffusion copes in two ways. It rescales the input, as EDM's $c_{\text{in}}(\sigma)\boldsymbol{u}$ does [4]. And it stops at a finite $\sigma_{\max}$, paying for the mismatch between $\mathcal{N}(\boldsymbol{0}, \sigma_{\max}^2 \boldsymbol{I})$ and $p_{\sigma_{\max}}$.

When the side information contains an image $\boldsymbol{x}_1$ close to $\boldsymbol{x}_0$, there is a better rescaling. Fix $V > 0$ and feed the network

$$\boldsymbol{x}_t = \boldsymbol{x}_1 + c(\sigma)\,(\boldsymbol{u} - \boldsymbol{x}_1), \qquad c(\sigma) = \frac{V}{\sigma^2 + V} = 1 - t, \qquad t = \frac{\sigma^2}{\sigma^2 + V} \in [0, 1]. \tag{4}$$

Proposition 1. *If $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$, then $\boldsymbol{x}_t = (1 - t)\,\boldsymbol{x}_0 + t\,\boldsymbol{x}_1 + \sigma_{\text{sb}}(t)\,\boldsymbol{\varepsilon}$ with $\sigma_{\text{sb}}(t)^2 = Vt(1 - t)$. This is the Brownian-bridge marginal from $\boldsymbol{x}_0$ to $\boldsymbol{x}_1$, with $V = 4\tau^2$. Conversely,*

$$\boldsymbol{u} = \frac{\boldsymbol{x}_t - t\,\boldsymbol{x}_1}{1 - t}, \qquad \sigma^2 = \frac{Vt}{1 - t}. \tag{5}$$

For fixed $\boldsymbol{x}_1$ and $t < 1$ the map is a bijection. So $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1] = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$: a regressor trained on bridge samples is *the denoiser that Algorithm 1 needs.*

Proof. Substitute: $\boldsymbol{x}_1 + (1 - t)(\boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon} - \boldsymbol{x}_1) = (1 - t)\boldsymbol{x}_0 + t\boldsymbol{x}_1 + (1 - t)\sigma\boldsymbol{\varepsilon}$, and $(1 - t)^2\sigma^2 = (1 - t)^2\, Vt/(1 - t) = Vt(1 - t)$. □

The level $\sigma = \sigma_{\text{sb}}/(1 - t) = \sqrt{V}\sqrt{t/(1 - t)}$ is a standard deviation. It diverges as $t \to 1$, although $\sigma_{\text{sb}} \to 0$, because σ_{sb} vanishes only like $\sqrt{1 - t}$. (The finite quantity $\sigma_{\text{sb}}^2/(1 - t) = Vt$ is the variance of the forward process before it is pinned at $\boldsymbol{x}_1$.)

Why it conditions the problem better. The bridge sample and the noisy target carry the same information, but the bridge sample is the much better network input:

- **Bounded noise.** The noise the network sees, $c(\sigma)\sigma = \sigma_{\mathrm{sb}}(t)$, peaks at τ (at $t = \frac{1}{2}$) and vanishes at both ends, while σ runs from 0 to ∞ (Figure 1).
- **Bounded signal, exact start.** The input stays near the segment from $\boldsymbol{x}_0$ to $\boldsymbol{x}_1$. At $\sigma = \infty$ it is exactly $\boldsymbol{x}_1$, a finite input the network is trained on. So the top of the ladder is $\sigma = \infty$ with no start mismatch, and the first step of Algorithm 1 is the network's regression $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$.
- **A Wiener-filter preconditioner.** (4) is $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}]$ under the Gaussian prior $\boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{x}_1, V\boldsymbol{I})$. So the network is fed the linear-Gaussian denoiser's output, and an $\boldsymbol{x}_0$-predicting network learns only the correction. The better $\boldsymbol{x}_1$ predicts $\boldsymbol{x}_0$, the smaller that correction.

The bridge parameter τ encodes an assumed per-pixel spread $\sqrt{V} = 2\tau$ of $\boldsymbol{x}_0 - \boldsymbol{x}_1$. It shapes the preconditioner and the ladder, but not the target: $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \sigma, \boldsymbol{x}_1]$ does not depend on V. A guide of a different contrast fits the Gaussian assumption worse. That makes the correction larger; it does not change what the network should learn.

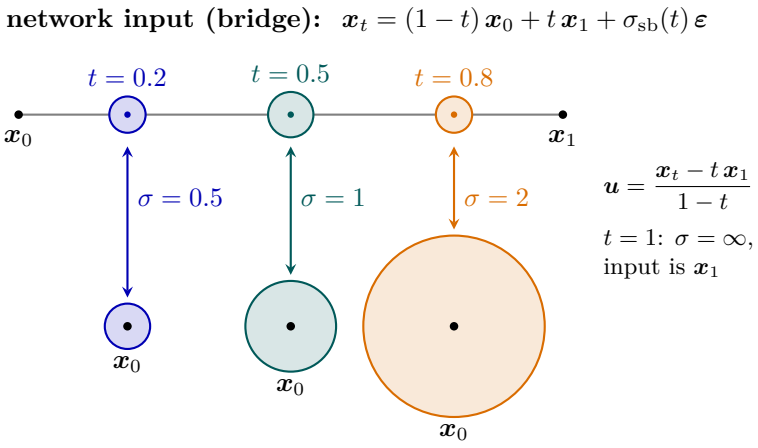


Figure 1: The same state in two coordinates. Bottom: the ascent state $\boldsymbol{u}$, the target $\boldsymbol{x}_0$ plus noise $\sigma(t)$, which grows without bound toward $t = 1$. Top: what the network is fed, the bridge sample (4), whose noise $\sigma_{\mathrm{sb}}(t)$ stays below τ and vanishes at both ends. The optimal outputs agree, $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1] = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$. Radii drawn to scale for $V = 1$.

Training. The I^{2}SB loss $\mathbb{E}\|D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) - \boldsymbol{x}_0\|^2$, with $t \sim \mathcal{U}[0,1]$, is conditional denoising with the preconditioner (4). In EDM's vocabulary, only the design choices differ from a standard conditional diffusion:

	conditional VE diffusion	bridge (I^{2}SB)
ideal denoiser	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \sigma, \boldsymbol{x}_1]$	the same
network input	$c_{\mathrm{in}}(\sigma)\,\boldsymbol{u}$, e.g. $(\sigma^2 + \sigma^2_{\mathrm{data}})^{-1/2}\boldsymbol{u}$	$\boldsymbol{x}_1 + c(\sigma)(\boldsymbol{u} - \boldsymbol{x}_1)$
training levels	e.g. log-normal σ	$p(\sigma) = 2\sigma V/(\sigma^2 + V)^2$
top level	finite $\sigma_{\max}$, start $\mathcal{N}(\boldsymbol{0}, \sigma^2_{\max}\boldsymbol{I})$	∞, start $\boldsymbol{x}_1$
sampling ladder	e.g. ρ-spaced in σ	uniform in t

The identity needs a network that is given $\boldsymbol{x}_1$. Without `cond_x1` (as in I^{2}SB-Inversion) it learns $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t]$. That network reads $\boldsymbol{x}_1$ only through $\boldsymbol{x}_t$, which works near $t = 1$ but is not the conditional denoiser in general.

3 I^{2}SB

3.1 The ascent I^{2}SB runs

Take Algorithm 1 with $\boldsymbol{c} = \boldsymbol{x}_1$ and three settings:

- the ladder $\sigma_k = \sigma(t_k)$ on a grid $1 = t_N > \cdots > t_0 = 0$, so $\sigma_N = \infty$;
- the ancestral step $h_k = 1 - r_k^2$;
- the denoiser read through Proposition 1, $D(\boldsymbol{u}, \sigma; \boldsymbol{x}_1) = D_\theta(\boldsymbol{x}_1 + c(\sigma)(\boldsymbol{u} - \boldsymbol{x}_1),\, t(\sigma),\, \boldsymbol{x}_1)$.

Write $t = t_k$, $s = t_{k-1}$. By (5), $r^2 = \sigma_s^2/\sigma_t^2 = \frac{s(1-t)}{t(1-s)}$, and the step is

$$\boldsymbol{u}_s = r^2\, \boldsymbol{u}_t + (1 - r^2)\, \boldsymbol{v} + \sigma_s \sqrt{1 - r^2}\; \boldsymbol{z}, \qquad \boldsymbol{v} = D(\boldsymbol{u}_t, \sigma_t; \boldsymbol{x}_1). \tag{6}$$

3.2 Line by line into I^{2}SB

Map every state to the network's coordinates with (4), $\boldsymbol{x}_t = (1 - t)\, \boldsymbol{u}_t + t\, \boldsymbol{x}_1$:

line	ascent in $\boldsymbol{u}$ (Algorithm 1)	I^{2}SB in $\boldsymbol{x}$
start	$\sigma_N = \infty$, $\boldsymbol{u}_N$ arbitrary	$\boldsymbol{x}_{t_N} = \boldsymbol{x}_1$
denoise	$\boldsymbol{v} = D(\boldsymbol{u}_t, \sigma_t; \boldsymbol{x}_1)$	$\hat{\boldsymbol{x}}_0 = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1)$
step	$\boldsymbol{u}_s = r^2 \boldsymbol{u}_t + (1 - r^2)\boldsymbol{v} + \sigma_s \sqrt{1 - r^2}\, \boldsymbol{z}$	$\boldsymbol{x}_s = \frac{s}{t}\boldsymbol{x}_t + \big(1 - \frac{s}{t}\big)\hat{\boldsymbol{x}}_0 + \sqrt{V \frac{s(t-s)}{t}}\, \boldsymbol{z}$
return	$\boldsymbol{u}_0 = \boldsymbol{v}$ at $\sigma_0 = 0$	$\boldsymbol{x}_0 = \hat{\boldsymbol{x}}_0$ at $t_0 = 0$

Proposition 2. *Each row of the table is the same operation in two coordinates. The right column is I^{2}SB's reverse step: the DDPM posterior of the bridge,* $\boldsymbol{x}_s \sim \mathcal{N}\big(\frac{s}{t}\boldsymbol{x}_t + (1 - \frac{s}{t})\hat{\boldsymbol{x}}_0,\; V \frac{s(t-s)}{t} \boldsymbol{I}\big)$.

Proof. Start. At $t = 1$, $c = 0$, so $\boldsymbol{x}_t = \boldsymbol{x}_1$ for every $\boldsymbol{u}_N$. *Denoise.* $\boldsymbol{x}_t$ is the input (4) of $\boldsymbol{u}_t$, so $\boldsymbol{v} = \hat{\boldsymbol{x}}_0$ by Proposition 1. *Step.* Multiply (6) by $1 - s$ and add $s\boldsymbol{x}_1$. The coefficients are

$$(1 - s)\, r^2 = \tfrac{s(1-t)}{t}, \qquad (1 - s)(1 - r^2) = \tfrac{t-s}{t}.$$

Substituting $\boldsymbol{u}_t = (\boldsymbol{x}_t - t\boldsymbol{x}_1)/(1 - t)$, the mean is

$$\tfrac{s}{t}(\boldsymbol{x}_t - t\boldsymbol{x}_1) + \big(1 - \tfrac{s}{t}\big)\hat{\boldsymbol{x}}_0 + s\boldsymbol{x}_1 = \tfrac{s}{t}\boldsymbol{x}_t + \big(1 - \tfrac{s}{t}\big)\hat{\boldsymbol{x}}_0.$$

The noise variance is

$$(1 - s)^2 \sigma_s^2 (1 - r^2) = (1 - s)^2 \cdot \tfrac{Vs}{1-s} \cdot \tfrac{t-s}{t(1-s)} = V \tfrac{s(t-s)}{t}.$$

Return. At $t_0 = 0$, $\boldsymbol{x} = \boldsymbol{u}$. □

For a general I^{2}SB schedule, replace t by $m_n = \sigma_{\text{fwd}}[n]^2/\sigma_{\text{fwd}}[N]^2$, and use $\sigma_n^2 = \sigma_{\text{fwd}}[N]^2\, m_n/(1 - m_n)$; the proof is unchanged. `debiased_ascent` (in `sb/i2sb_recon.py`) implements the left column. With I^{2}SB's step size and seed it reproduces `reverse_sample` iterate for iterate, to $2 \cdot 10^{-5}$ relative or better, with and without the CG step.

3.3 What I^{2}SB fixes, and what it frees

- **The warm start is the first step.** With $\sigma_N = \infty$, the first update is $\boldsymbol{u}_{t_{N-1}} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1] + \sigma_{N-1}\boldsymbol{z}$: regress, re-noise, then diffuse. This is the structure of SDEdit [5] and come-closer-diffuse-faster [6], with the initialisation from the network's own regression.
- τ **sets the noise scale.** On a fixed t-grid every σ_k scales with $\sqrt{V} = 2\tau$. At 20 uniform steps ($t_{N-1} = 0.95$), $\sigma_{N-1} \approx 1.7$ for $\tau = 0.19$ and ≈ 0.8 for $\tau = 0.09$, against image intensities ~ 1.
- **The start image** (zero-filled, ACS, guide) changes only the conditioning and the centre of the preconditioner.
- **Freed by the $\boldsymbol{u}$-form.** Algorithm 1 accepts the same trained network with any step $h \in [1-r, 1]$ (for example the deterministic $h = 1 - r$), any ladder, or a finite-σ start from noise. I^{2}SB fixes $h = 1 - r^2$, the uniform t-grid and $\sigma_N = \infty$.

4 With a measurement: I^{2}SB-Recon is ImMAP2

Adding a measurement $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x}_0 + \boldsymbol{n}$, with $\boldsymbol{n} \sim \mathcal{N}(\boldsymbol{0}, \sigma_n^2 \boldsymbol{I})$, to the side information changes only the target, $\boldsymbol{c} = (\boldsymbol{x}_1, \boldsymbol{y})$. Since $\boldsymbol{\varepsilon}$ is independent of $\boldsymbol{y}$, (1) still holds with $D = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$. The network supplies $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$. Approximating $p(\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1) \approx \mathcal{N}(\hat{\boldsymbol{x}}_0, \varsigma^2 \boldsymbol{I})$ folds the data in:

$$\bar{\boldsymbol{x}}_0 = (\boldsymbol{E}^H \boldsymbol{E} + \lambda \boldsymbol{I})^{-1} (\boldsymbol{E}^H \boldsymbol{y} + \lambda \hat{\boldsymbol{x}}_0) \approx \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}], \qquad \lambda = \sigma_n^2 / \varsigma^2. \tag{7}$$

This is the CG step of I^{2}SB-Recon [7]. ImMAP2 (`diffusion/immap.py`) runs, at level σ_k: denoise to $\hat{\boldsymbol{x}}_0$, apply (7), then

$$\boldsymbol{u}_{k+1} = \boldsymbol{u}_k + h_k(\bar{\boldsymbol{x}}_0 - \boldsymbol{u}_k) + \sigma_k \sqrt{(1 - \beta h_k)^2 - (1 - h_k)^2}\, \boldsymbol{z}, \qquad \sigma_{k+1} = (1 - \beta h_k)\, \sigma_k. \tag{8}$$

Here σ_{k+1} is the level the injection aims for; ImMAP2 then re-estimates it from the next residual. This is Algorithm 1 with $r = 1 - \beta h$.

Proposition 3. *With $r = \sigma_{k+1}/\sigma_k$, $h_k = 1 - r^2$ and $\beta = 1/(1+r)$, the ImMAP2 step (8) is the I^{2}SB step (6) with $\hat{\boldsymbol{x}}_0$ replaced by $\bar{\boldsymbol{x}}_0$. That is I^{2}SB-Recon, written in the coordinates (5).*

Proof. $1 - \beta h_k = (1 + r - 1 + r^2)/(1 + r) = r$, so the next level is σ_{k+1}. The mean $\boldsymbol{u}_k + (1 - r^2)(\bar{\boldsymbol{x}}_0 - \boldsymbol{u}_k)$ is that of (6), and the noise variance is $\sigma_k^2(r^2 - r^4) = \sigma_{k+1}^2(1 - r^2)$. □

So the two methods share the step structure exactly and differ only in its settings:

	ImMAP2	I^{2}SB-Recon
denoiser	unconditional, prior $p(\boldsymbol{x}_0)$	conditional, prior $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)$
network input	$\boldsymbol{u}$	bridge sample (4)
noise ladder	closed loop, $\sigma_k^2 = \frac{1}{n}\lVert\hat{\boldsymbol{x}}_0 - \boldsymbol{u}_k\rVert^2$	open loop, fixed t-grid
step size h_k	schedule $h_0 k/(1 + h_0(k-1))$	$1 - r^2$, the variance removed
β	fixed (0.05 by default)	$1/(1+r)$: 1 on the first step, $\to \frac{1}{2}$
prox weight λ	$\lambda_0 \sigma_n^2 (1 + \sigma_k^2)/\sigma_k^2$	constant $(7 \cdot 10^{-5})$
start	noise at $\sigma_{\max}$, or a warm start	$\boldsymbol{x}_1$ at $\sigma = \infty$: regress, re-noise

Modelling differs in the conditional prior and the bridge input; the rest are schedules.

- **The prox weight.** ImMAP2's λ_k is (7) with $\varsigma^2 = \sigma_k^2/(1 + \sigma_k^2)$, the posterior variance under a unit-variance Gaussian prior, so it follows the noise level. A constant λ assumes the denoiser is equally uncertain at every level.

- **The start decides what the prox contributes.** For the zero-filled start, $\boldsymbol{x}_1 = \boldsymbol{E}^H \boldsymbol{y}$ is sufficient for $\boldsymbol{y}$, so $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$ already uses the data and (7) only corrects the network's errors. For a guide start, (7) is the only way $\boldsymbol{y}$ enters, as in ImMAP2.

5 Relation to standard diffusion

Since $\boldsymbol{u}$ and $\boldsymbol{x}_1$ are independent given $\boldsymbol{x}_0$, Bayes' rule splits the conditional score, and Tweedie turns the split into denoisers:

$$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}] + \sigma^2 \nabla_{\boldsymbol{u}} \log p_\sigma(\boldsymbol{x}_1 \mid \boldsymbol{u}). \tag{9}$$

So I^{2}SB is an unconditional VE diffusion plus a guidance term for $\boldsymbol{x}_1$. That term is learned, where DPS [8] approximates it through the denoiser's Jacobian.

- **Classifier-free guidance** would need the unconditional denoiser in the same $\boldsymbol{u}$-coordinates. The bridge input depends on $\boldsymbol{x}_1$, so $D_\varnothing$ cannot share I^{2}SB's preconditioner.
- **A conditional-diffusion baseline** on the same $\boldsymbol{x}_1$ (SR3/Palette style) is not an independent method. It has the same target and the same sampler family, so any gap in results comes from the design choices of Section 2: above all, how well the network copes with its inputs at large σ.

References

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